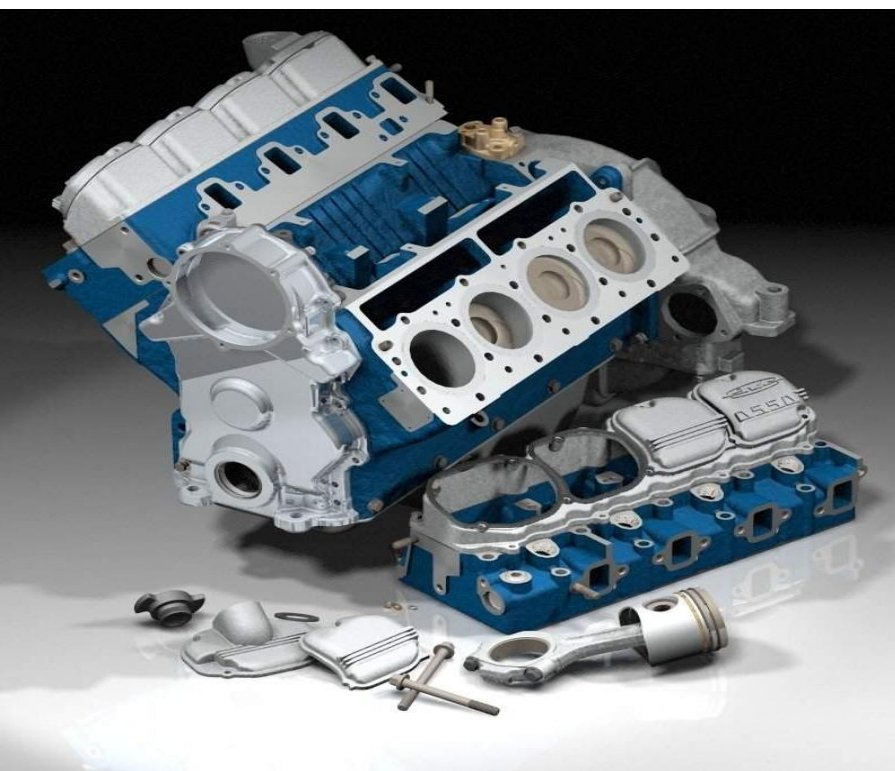


第一章 内燃机工作原理及总体构造

The Working Principles and Overall Structure of Internal Combustion Engines



第一节 汽车发动机的类型

Types of Automotive Engines

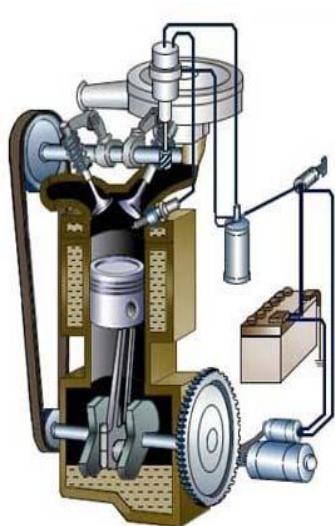
- **发动机**：将某一种形式的能量（热能、电能、化学能、太阳能等）转变成机械能的机器。
- **Engine** is a kind of machine which turns a certain kind of energy to mechanical energy .
- **热力发动机**：将热能转变成机械能的发动机。
- **Heat engine** is the kind of engine which changes thermal energy into mechanical energy .
- 热力发动机包括**内燃机**和外燃机。
- Heat engines include **internal combustion engine** and external combustion engine.
- 传统汽车发动机属于**往复活塞式内燃机**或者**转子活塞式发动机**。
- General automotive engines are **Reciprocating Piston Engines** or **Rotary Engines**.

汽车发动机的类型

Types of Automobile Engines

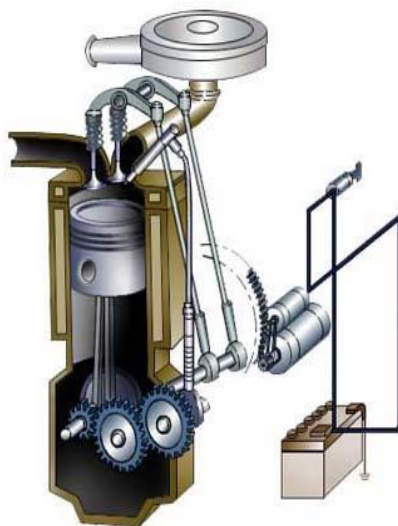
(1) 按照所用燃料分类

Classification by Fuel



汽油机

Gasoline engine

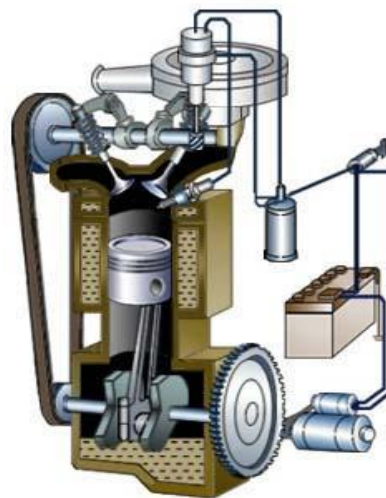


柴油机

Diesel engine

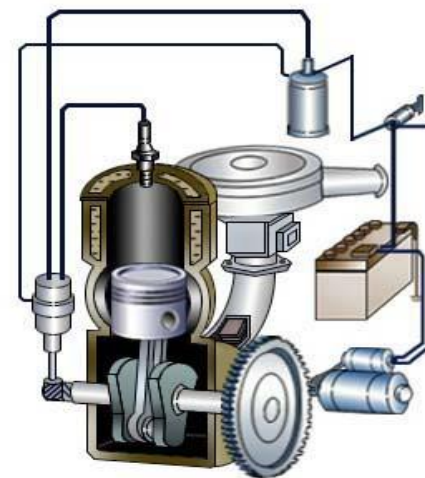
(2) 按照工作行程分类

Classification by Working Cycle



四行程

Four stroke cycle

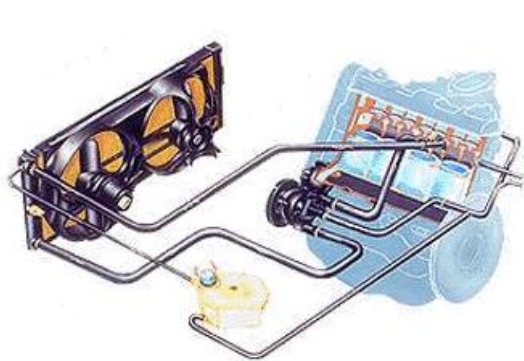


二行程

Two stroke cycle

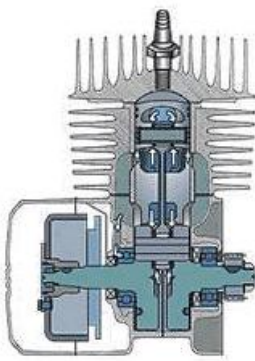
(3) 按照冷却方式分类

Classification by Cooling System



水冷

Water cooled



风冷

Air cooled

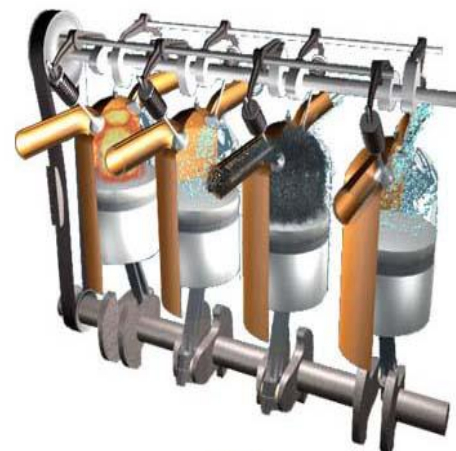
(4) 按照气缸数目分类

Classification by Cylinder Number



单缸

Single-cylinder

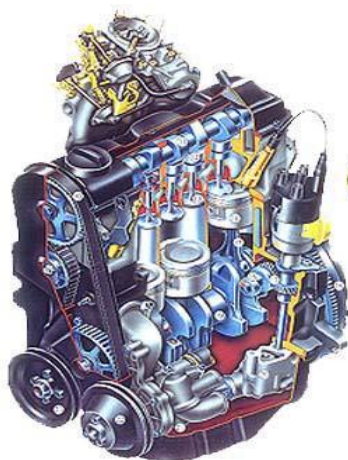


多缸

Multi-cylinder

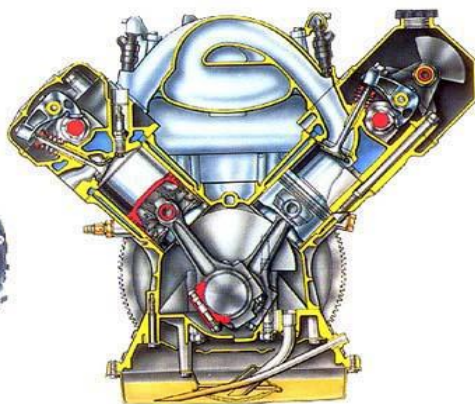
(5) 按照气缸排列方式分类

Classification by Cylinder Arrangement



直列

Inline engine



V 型

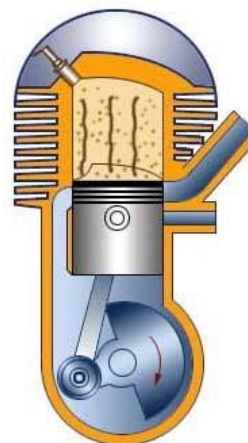
V-engine

除此以外，还有W型发动机和直列卧式发动机等等

Besides ,there are W-engine , Flat-opposed engine and so on.

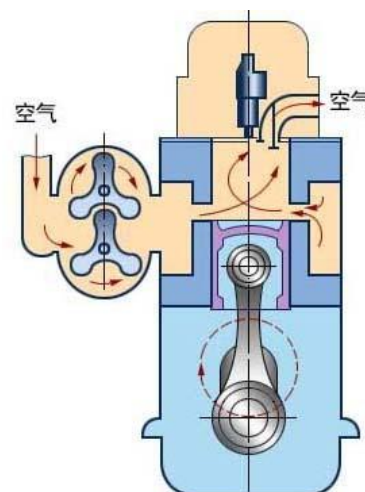
(6) 按照是否采用增压分类

Classification by Charging System



自然吸气

Naturally aspirated



增压

Supercharged
or Turbocharged

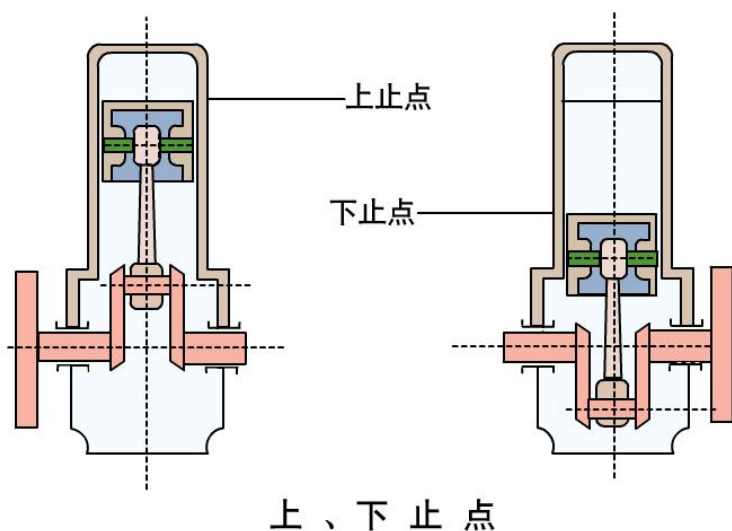
第二节 发动机的基本术语

Basic Concepts of Engines

(1) 上、下止点

Top dead centre/TDC

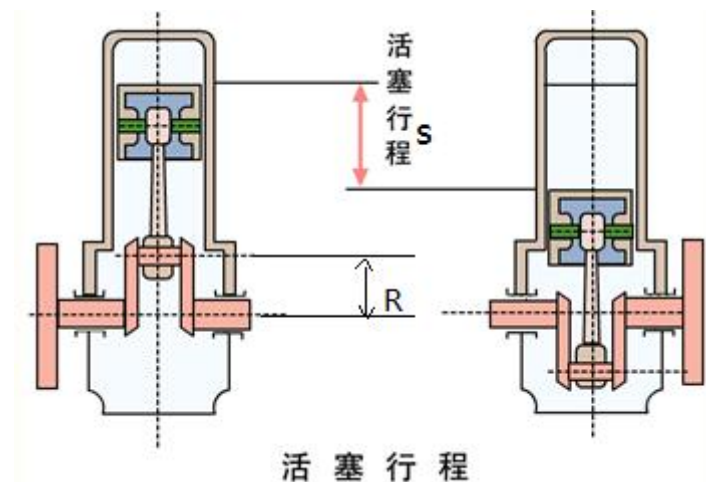
Bottom dead centre/BDC



(2) 活塞行程 Piston stroke S

曲柄半径 Crank radius R

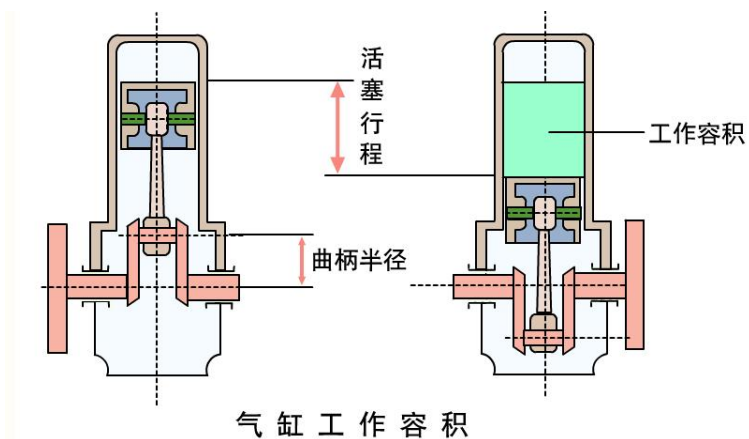
$$S=2R$$



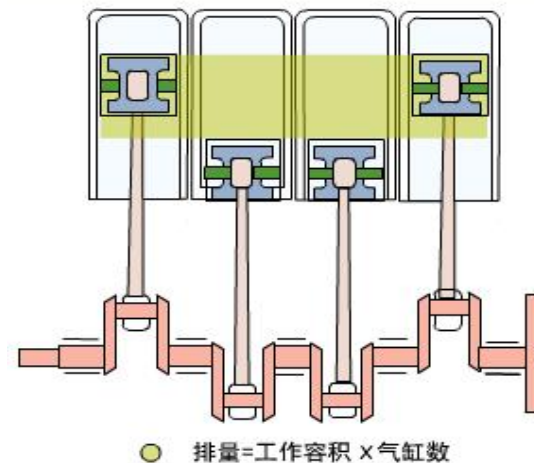
上、下止点间的距离 S 称为活塞行程。

The distance between top dead centre and bottom dead centre is called piston stroke.

(3) 气缸工作容积 Swept volume V_h

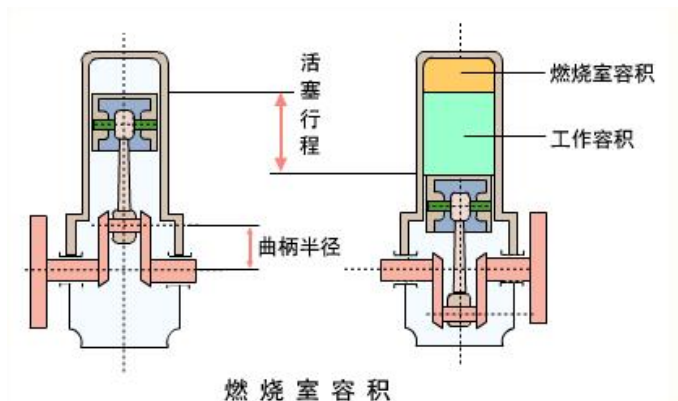


(4) 发动机排量 Displacement V_H

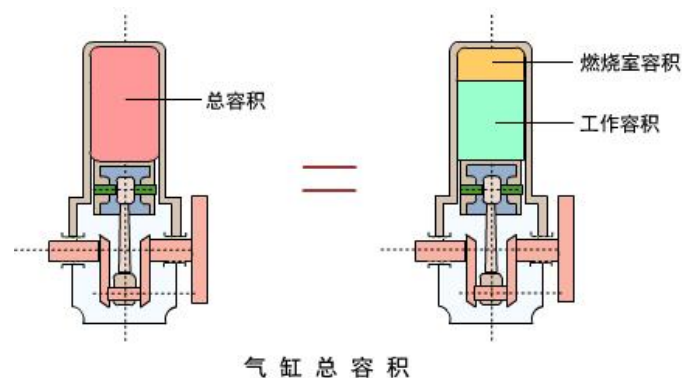


$$V_H = V_h \times Z \quad Z = \text{number of cylinders}$$

(5) 燃烧室容积 Clearance volume V_c

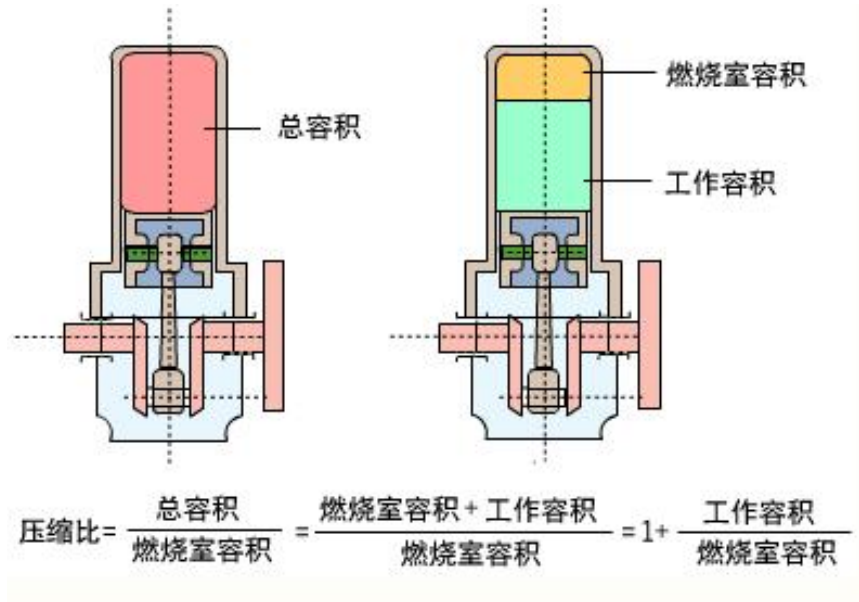


(6) 气缸总容积 cylinder all volume V_a



$$V_a = V_h + V_c$$

(7) 压缩比 Compression Ratio ε



$$\varepsilon = \frac{V_a}{V_c} = \frac{V_h + V_c}{V_c} = 1 + \frac{V_h}{V_c}$$

压缩比是发动机中一个非常重要的概念，压缩比表示了气体的压缩程度，它是气体压缩前的容积与气体压缩后的容积之比值，即气缸总容积与燃烧室容积之比称为压缩比。一般用 ε 表示。

The compression ratio is an important concept about engines. It shows the compression degree of gas and is defined as the quotient of the maximum and minimum cylinder volumes.

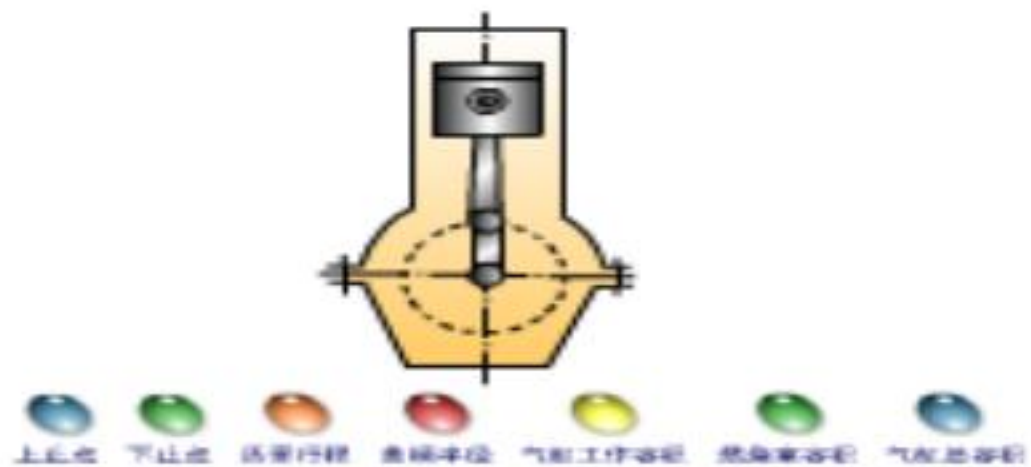
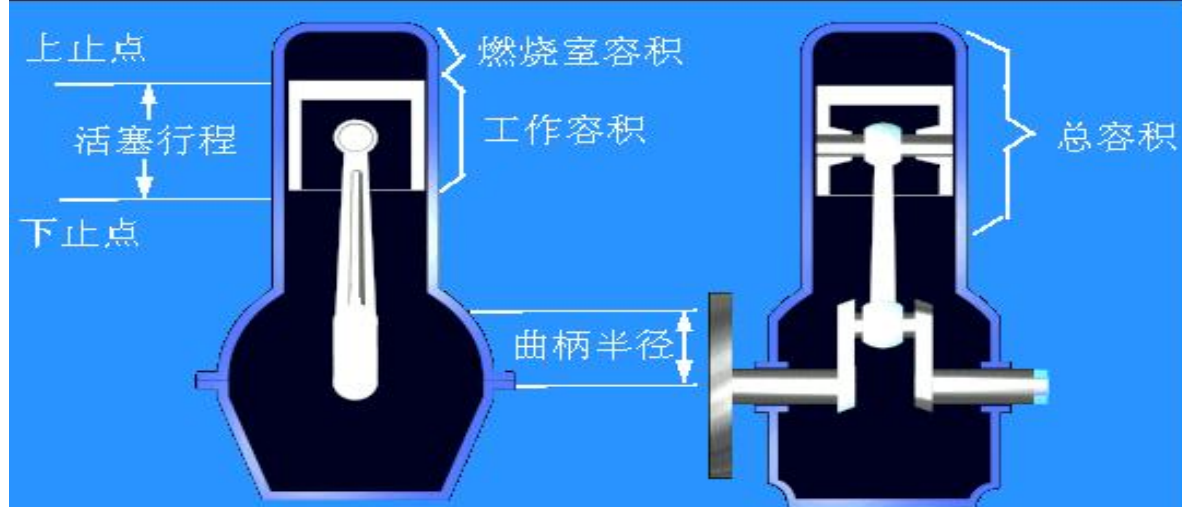


图1-1 发动机基本术语动画演示

第三节 往复式活塞式内燃机工作原理

The Working Principles of Reciprocating Piston Engines

一、四冲程内燃机工作原理 Four-Stroke Engine

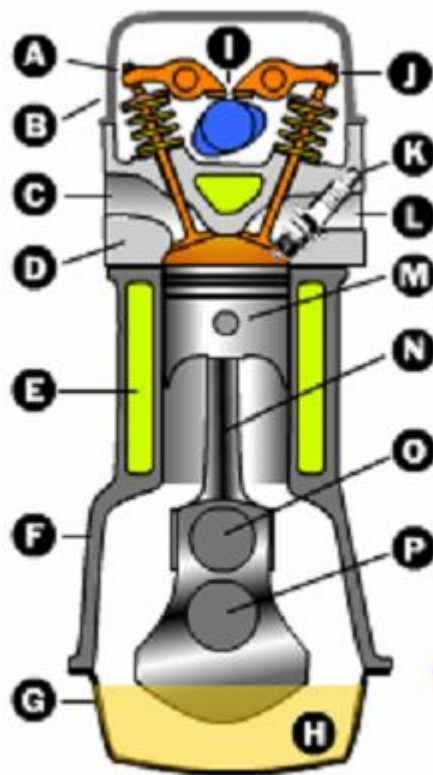
内燃机基本工作原理就是可燃混合气在气缸内燃烧放热并通过曲柄连杆机构对外做功。汽车发动机在气缸内产生能量的过程叫燃烧周期。大多数发动机的燃烧周期分为四个过程（也叫四冲程），也就是说，在一个做功周期内活塞上下运动两次。为了纪念德国工程师尼古拉斯·奥托首创这种四冲程做功方式，也把称它为奥托循环。

The basic working principle of IC engine is combustible mixed gas burns in cylinder and work through crank and connecting rod system. The process in which automotive engines produce energy within the cylinder is called combustion cycle. Most of engines includes four periods (four stroke) within one combustion cycle. That is, the piston moves up and down twice in a working cycle. This four-stroke working mode was first developed by the German engineer Nicolaus Otto, so it is called Otto cycle as well.

下面我们分别来介绍一下四冲程汽油机和四冲程柴油机的工作原理和工作过程：

Now let's introduce the working principles and working processes of four-stroke gasoline engines and four-stroke diesel engines :

四冲程汽油机工作原理 Four-Stroke Gasoline Engine



©2000 How Stuff Works, Inc.

- | | |
|--|---|
| A Intake Valve, Rocker Arm & Spring | I Camshaft |
| B Valve Cover | J Exhaust Valve, Rocker Arm & Spring |
| C Intake port | K Spark Plug |
| D Head | L Exhaust Port |
| E Coolant | M Piston |
| F Engine Block | N Connecting Rod |
| G Oil Pan | O Rod Bearing |
| H Oil Sump | P Crankshaft |

A进气阀、摇臂、弹簧
B气门盖
C进气口
D气缸盖
E冷却液
G油底壳
I曲轴
J排气阀、摇臂、弹簧
K火花塞
L排气口
M活塞
N连杆
O连杆轴承
P曲轴

- | | |
|------------------------|--------|
| 1 INTAKE | ① 进气行程 |
| 2 COMPRESSION | ② 压缩行程 |
| 3 COMBUSTION | ③ 做功行程 |
| 4 EXHAUST | ④ 排气行程 |
| Spark | |
| Top Dead Center | |

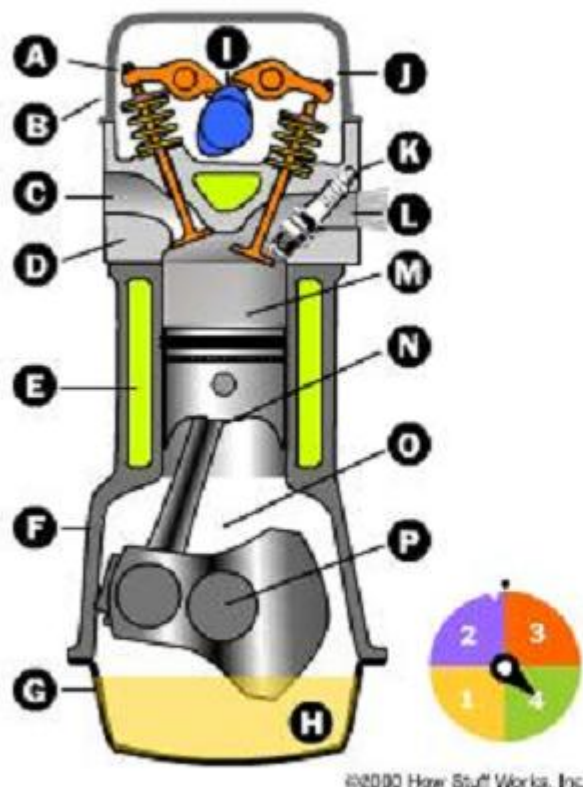


Inside a Typical Gasoline Engine



四冲程汽油机工作原理动画演示

Schematic Illustration of Four-Stroke Gasoline Engine



Inside a Typical Gasoline Engine

It should be noted:

Every 2 turns of the engine crankshaft complete a working cycle in a 4-stroke engine.

For producing a higher working efficiency, the followings should be paid more attention:

进气门开 IVO --- Intake Valve Open

进气门关 IVC --- Intake Valve Close

排气门开 EVO ---Exhaust Valve Open

排气门关 EVC --- Exhaust Valve Close

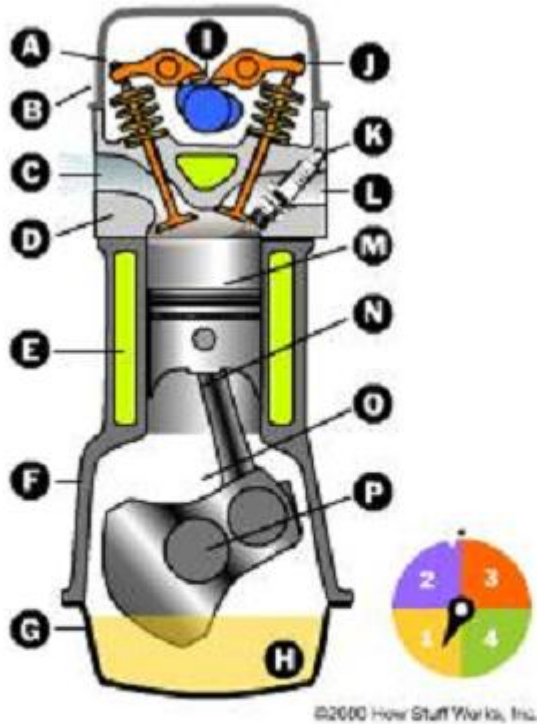
点火时刻 Ignition Timing



进气行程

The Intake Stroke:

1. Piston moves down
2. The intake valve has opened.
3. A mixture of air and fuel enters into the cylinder through the intake valve port



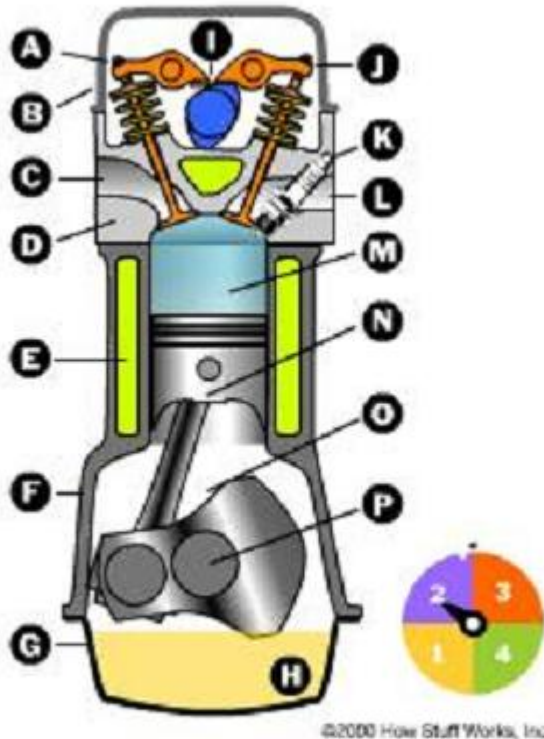
IVO too late and IVC too early, a few of fresh mixture is induced into the cylinder.

Intake Stroke of Gasoline Engine

压缩行程

The Compression Stroke:

1. After the piston reaches the lower limit of its travel in cylinder (BDC)
2. The piston begins to move upward.
3. As this happens, the intake valve closes.
4. The exhaust valve is also closed, so the cylinder is sealed.
5. As the piston moves upward, the air/fuel mixture is compressed.

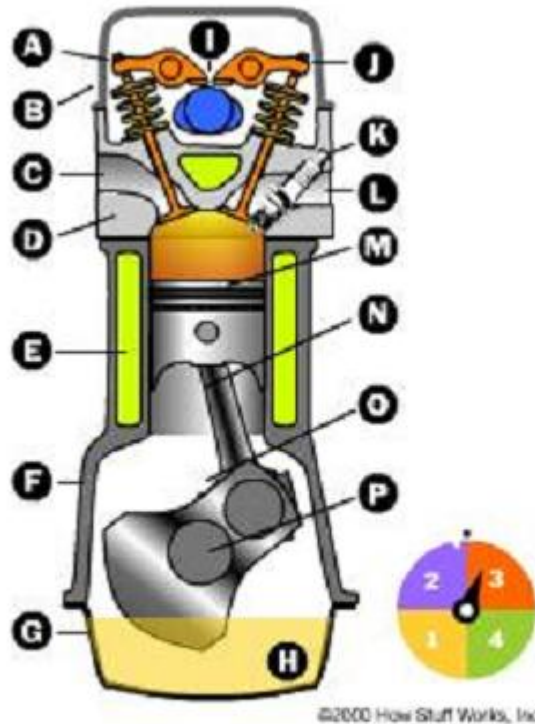


Compression Stroke of Gasoline Engine

做功行程

Combustion Stroke (Expansion):

1. As the piston reaches near to the top of its travel in the cylinder (TDC) on the compression stroke, the spark plug delivers a spark to.
2. The spark ignites the air/fuel mixture.
3. The burnt gas produces a high pressure on the piston and forces the piston down .



Combustion Stroke of Gasoline Engine

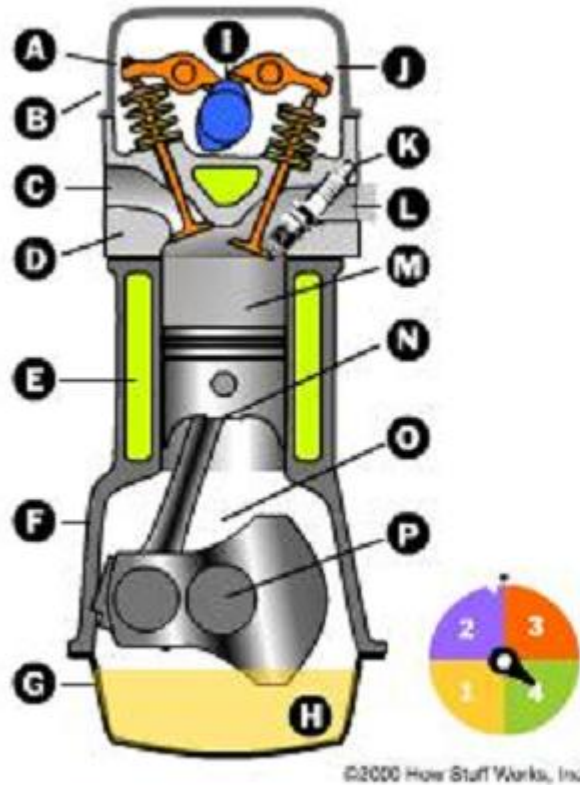
**Ignite too late, produce less power.
And directly influence the pollution
formation**

**Ignite too early, produce knock
combustion.**

排气行程

The Exhaust Stroke:

1. As the piston reaches the bottom of its travel (BDC), the exhaust valve opens.
2. Now, as the piston moves up on the exhaust stroke, it forces the burned gases out of the cylinder through the exhaust port.
3. When the piston reaches the top of its travel (TDC), the exhaust valve closes, and the intake valve opens.

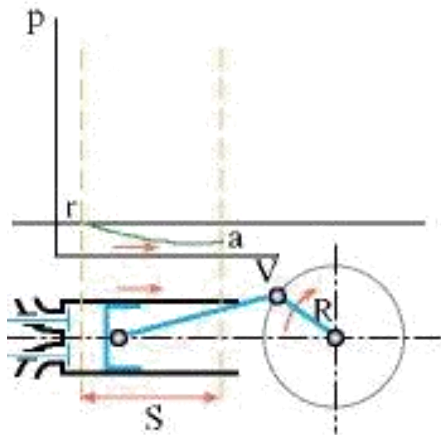


Exhaust Stroke of Gasoline Engine

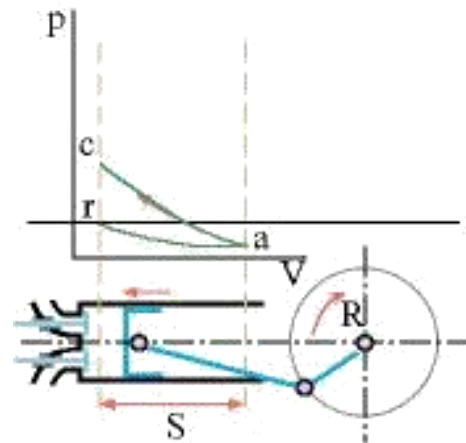
EVO too early, waste useful work.

EVO too late, unable to take out exhaust gas.

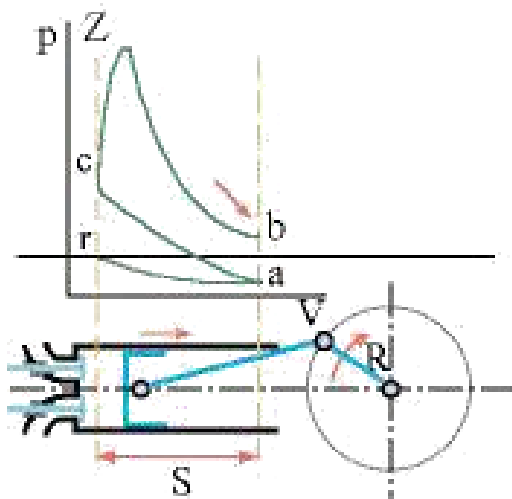
发动机工作循环 Working Cycle of Engine



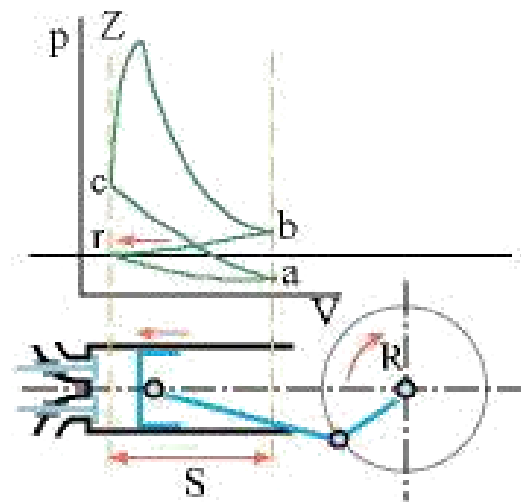
进气行程
Intake



压缩行程
Compression



做功行程
Power

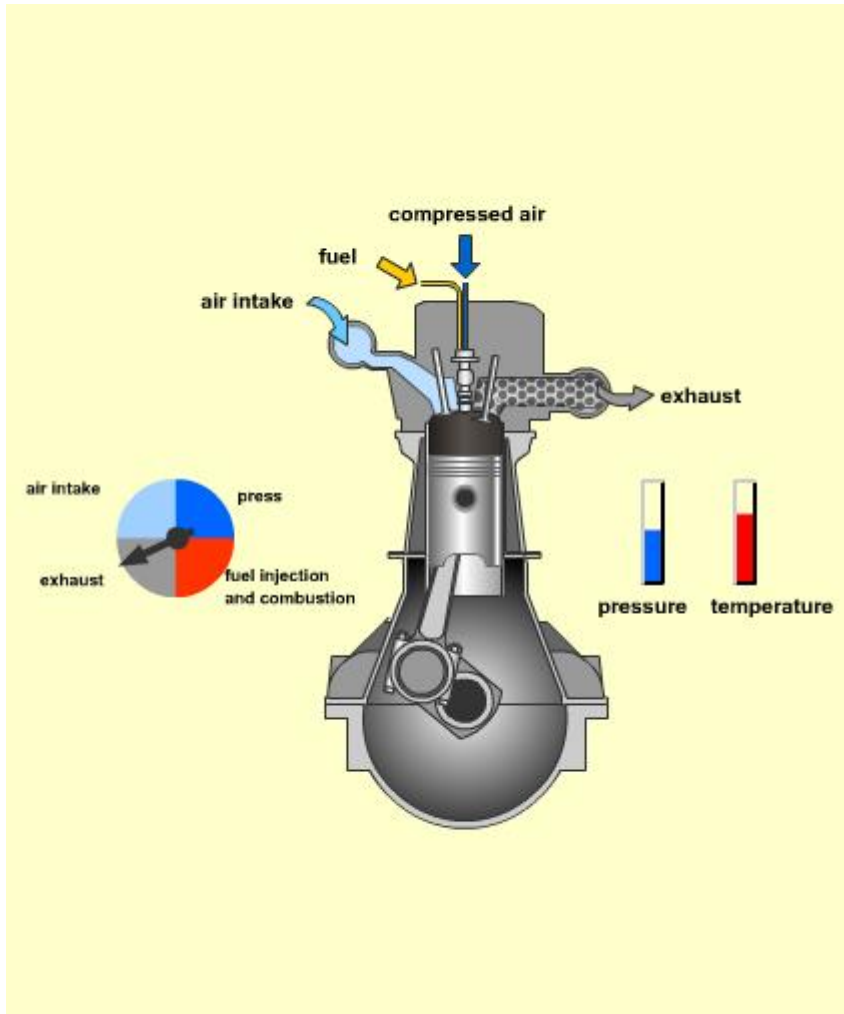


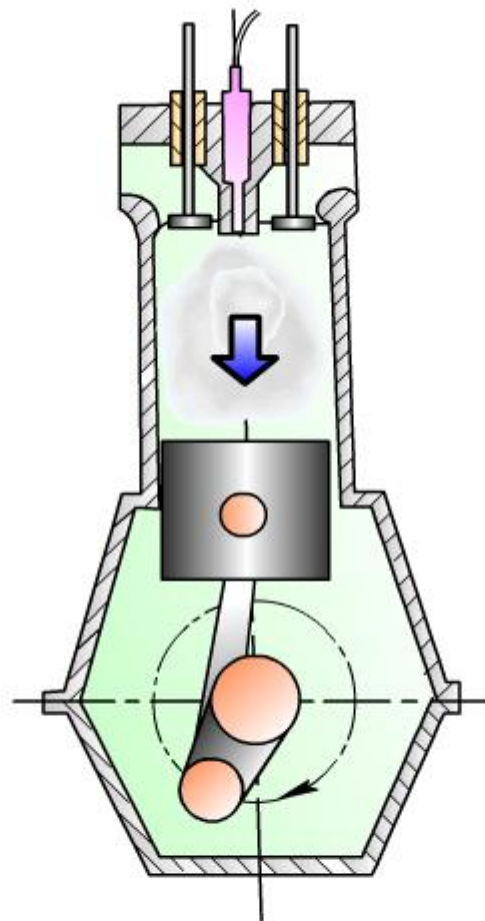
排气行程
Exhaust

四冲程柴油机的工作原理 Four-Stroke Diesel Engine

四行程柴油机和四行程汽油机的工作过程相同，每一个工作循环同样包括进气、压缩、作功和排气四个行程，但由于柴油机使用的燃料是柴油，柴油与汽油有较大的差别，柴油粘度大，不易蒸发，自燃温度低，故可燃混合气的形成，着火方式，燃烧过程以及气体温度压力的变化都和汽油机不同，下面主要分析一下柴油机和汽油机在工作过程中的不同点。

The working process of four-stroke diesel engine is same with the Otto engine. There are also four stroke in one working cycle. However, diesel engine uses diesel as fuel, and diesel is quite different with gasoline for diesel has bigger viscosity, is difficult to evaporate, and gets low ignition temperature, so the mixture formation, ignition, combustion process and the change of gas temperature and pressure are different.





● 进气行程 ● 压缩行程 ● 作功行程 ● 排气行程 ● 连续演示

图1-3 单缸四冲程柴油机工作原理示意图

Schematic Illustration of Four-Stroke Diesel Engine

汽油机和柴油机的不同点

The differences between Gasoline engine and Diesel engine

(1)汽油机的可燃混合气在缸外部开始形成并延续到进气和压缩行程终了，时间较长。柴油机的可燃混合气在气缸内部形成，从压缩行程接近终了时开始，并占小部分作功行程，时间很短。

(1) In an Otto engine, the air-fuel mixture starts to form out of cylinder and continues until the intake stroke and compression stroke finish. In a diesel engine , the air-fuel mixture forms in the cylinder, starts from near the end of the compression stroke , and accounts for a small part of combustion stroke. The period is short.

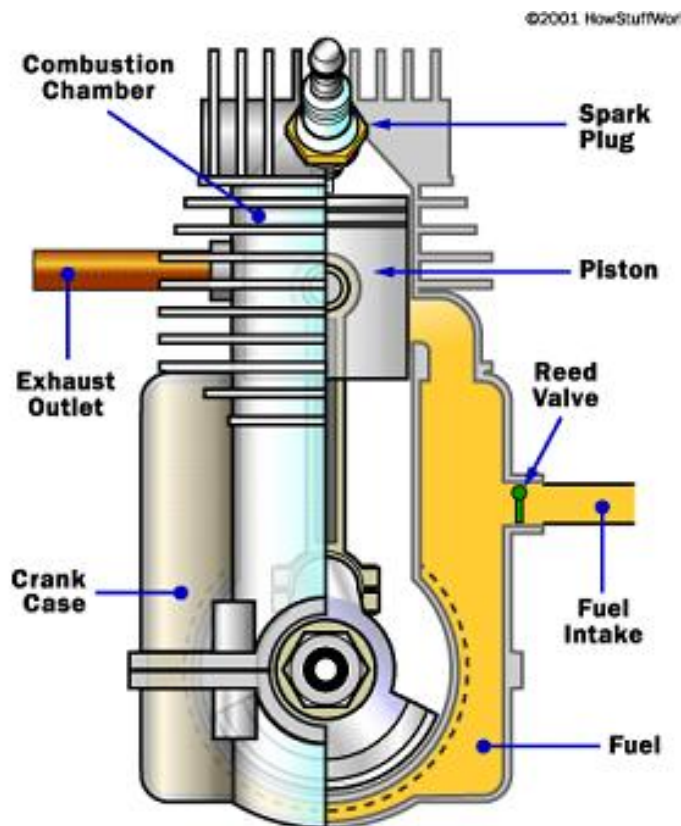
(2)汽油机的可燃混合气用电火花点燃，柴油机则是自燃。所以又称汽油机为点燃式内燃机，称柴油机为压燃式内燃机。

(2) In an Otto engine the spark ignites the air-fuel mixture. However in a diesel engine, the mixture ignites spontaneously without a spark. So the diesel engine is called compression ignition engine/CI engine.

二、二冲程内燃机工作原理

Working Principles of Two-Stroke Engine

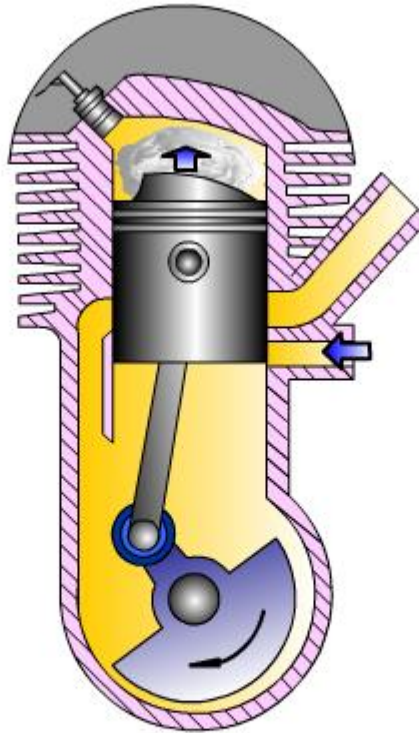
二冲程汽油机工作原理 Two Stroke SI Engine



The basic components of a two-stroke engine

二行程汽油机的工作循环也是由进气、压缩、燃烧膨胀、排气过程组成，但它是在曲轴旋转一圈(360°)，活塞上下往复运动的两个行程内完成的。因此，二行程发动机与四行程发动机工作原理不同，结构也不一样。

Two-stroke engine does require 4 processes to complete a working cycle, including Intake, Compression, Combustion and Exhaust. It just needs one rotation of engine crankshaft to complete a working cycle with two stroke of piston. So the working principles and structure between two-stroke engine and four-stroke engine are both different.

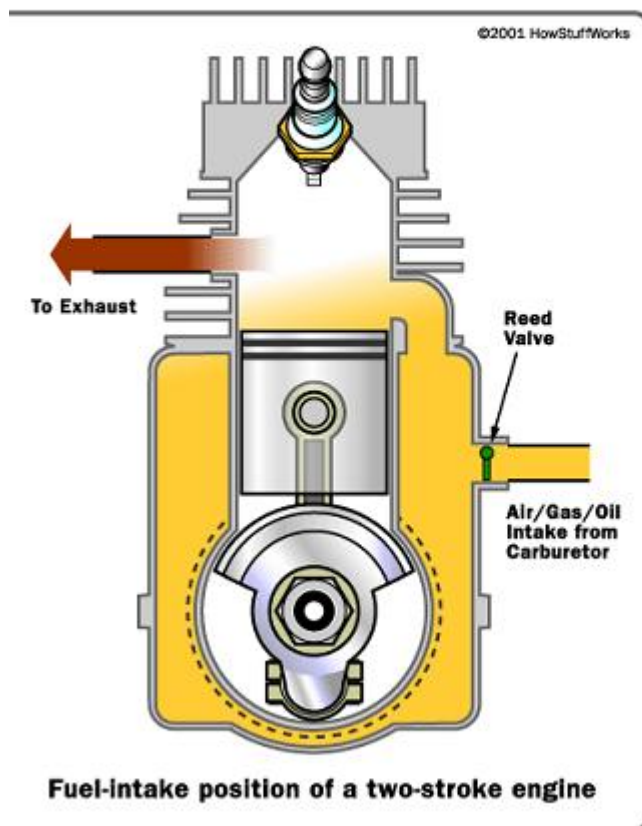


● 进气行程 ● 压缩行程 ● 做功行程 ● 排气行程 ● 连续演示

二冲程汽油机工作原理示意图

Schematic Illustration of Two-Stroke Gasoline Engine

Two Stroke SI Engine



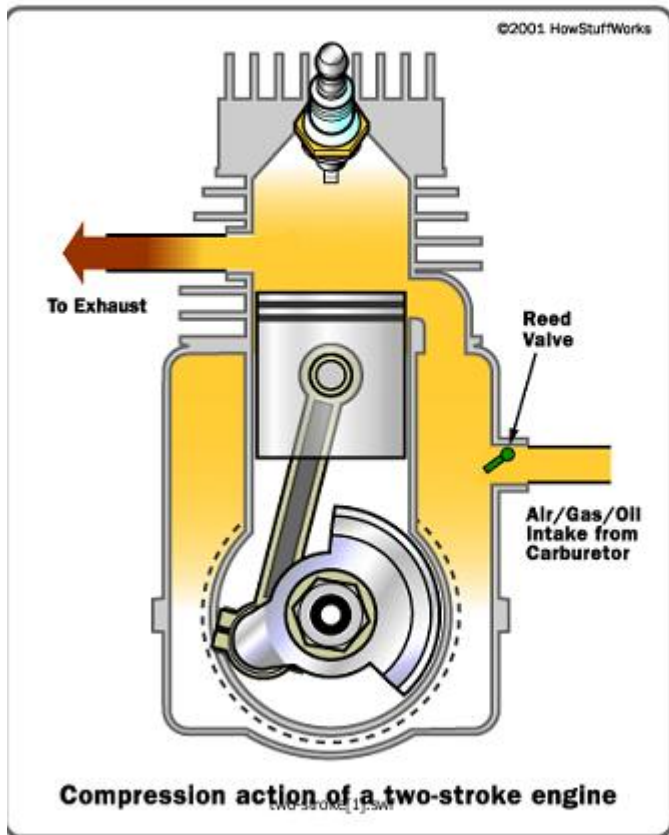
Fuel Intake Process

As the piston bottoms out, the **intake port** is uncovered.

The piston's movement when piston moved down has **pressurized** the mixture in the crankcase, so it rushes into the cylinder, **displacing** the remaining exhaust gases and filling the cylinder with a fresh charge of fuel/air mixture.

Note that in many two-stroke engines that use a **cross-flow design on piston top**, the piston is shaped so that the incoming fuel mixture doesn't simply flow right over the top of the piston and out the exhaust port.

Two Stroke SI Engine



Compression Stroke

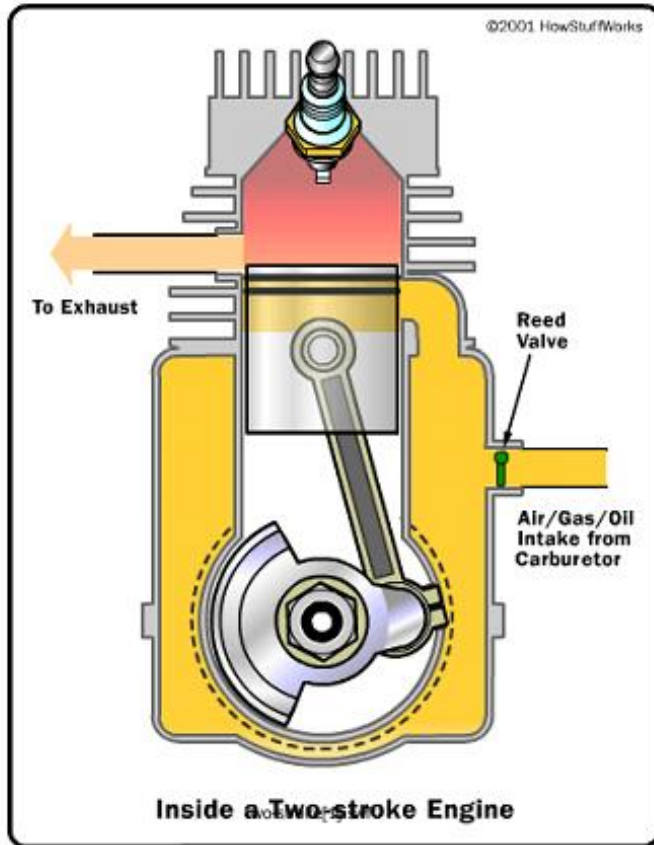
Now the piston starts driving the piston back up toward the spark plug for the **compression stroke**.

As the air/fuel mixture in the piston is compressed, a **vacuum** is created in the crankcase.

This vacuum opens the **reed valve** and sucks air/fuel/oil in from the **carburettor**.

Piston moves up, covers the exhaust port and presses the mixture continually until the spark ignites the mixture.

Two Stroke SI Engine



Combustion Stroke and Exhaust Process

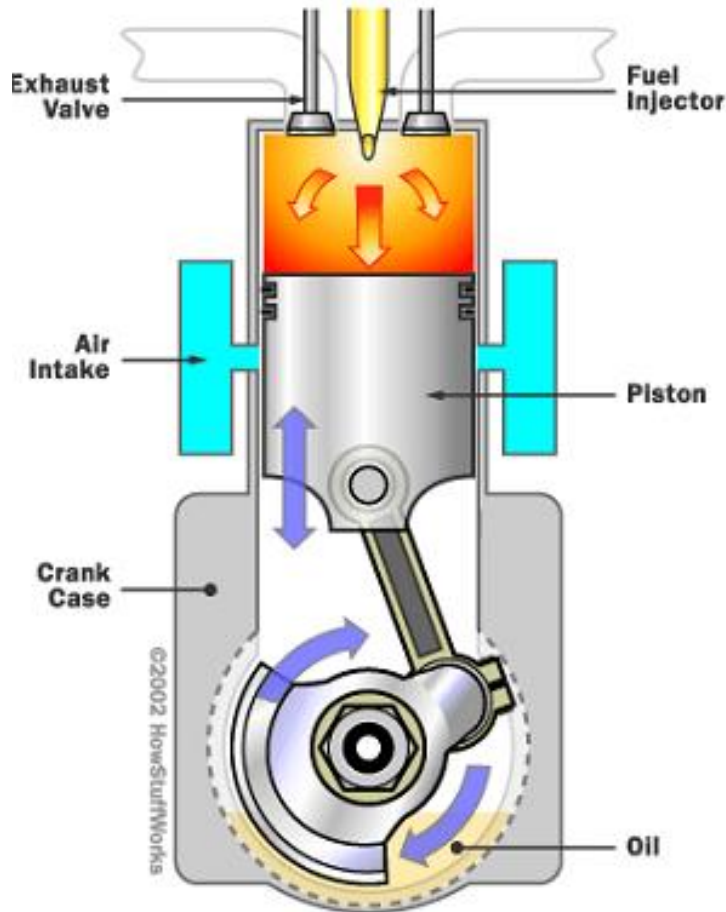
The **spark plug** fires compressed mixture around the TDC.

The resulting **explosion** drives the **piston** downward.

As the piston moves down, the **exhaust port** is uncovered. The **pressure** in the cylinder drives most of the exhaust gases out of cylinder until the intake port uncovers.

Note that as the piston moves downward, it is compressing the air/fuel mixture in the crankcase.

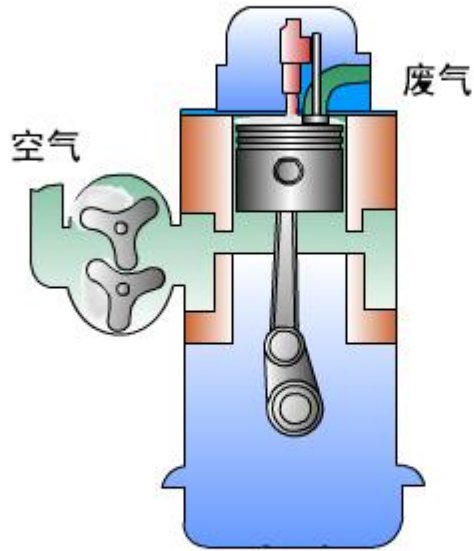
二冲程柴油机工作原理 Two Stroke Diesel Engine



二行程柴油机和二行程汽油机工作类似，所不同的是，柴油机进入气缸的不是可燃混合气，而是纯空气。例如带有扫气泵的二行程柴油机工作过程如下：

第一行程：活塞从下止点向上止点运动，行程开始前不久，进气孔和排气门均以开启，利用从扫气泵流出的空气使气缸换气。当活塞继续向上运动进气孔被关闭，排气门也关闭，空气受到压缩，当活塞接近上止点时，喷油器将高压柴油以雾状喷入燃烧室，燃油和空气混合后燃烧，使气缸内压力增大。

第二行程：活塞从上止点向下止点运动，开始时气体膨胀，推动活塞向下运动，对外做功，当活塞下行到大约2/3行程时，排气门开启，排出废气，气缸内压力降低，进气孔开启，进行换气，换气一直延续到活塞向上运动1/3行程进气孔关闭结束。



单缸二冲程柴油机工作原理

Working Principles of Single-Cylinder Two-Stroke Diesel Engine

The two-stroke diesel cycle works like this:

Power stroke: when the piston is near to the top of its travel, the cylinder contains a highly compressed air, diesel fuel is sprayed into the cylinder by the injector and immediately ignites because of the heat and pressure inside the cylinder. The pressure created by the combustion of the fuel drives the piston downward.

Intake and exhaust process: the piston nears the bottom of its travel, all of the exhaust valves open. Exhaust gases rush out of the cylinder, relieving the pressure. And then the piston bottoms out, it uncovers the air intake ports. Pressurized air fills the cylinder, forcing out the remainder of the exhaust gases.

Compression Stroke: the exhaust valves close and the piston starts travelling back upward, recovering the intake ports and compressing the fresh charge of air.

四冲程和二冲程柴油机的比较 [动画](#)

Comparison of the four-stroke and two-stroke diesel engine

四冲程和二冲程汽油机的比较 [动画](#)

Comparison of the four-stroke and two-stroke gasoline engine

Two-stroke engines have three important advantages over four-stroke engines:

1. Two-stroke engines do not have valves, which simplifies their construction and lowers their weight.
2. Two-stroke engines fire **once every revolution**, while four-stroke engines fire **once every other revolution**. This gives two-stroke engines a significant power boost.
3. Two-stroke engines can work in any orientation, which can be important in something like a chainsaw. A standard four-stroke engine may have problems with oil flow unless it is upright, and solving this problem can add complexity to the engine.

These advantages make two-stroke engines lighter, simpler and less expensive to manufacture compared to many four-stroke engine designs.

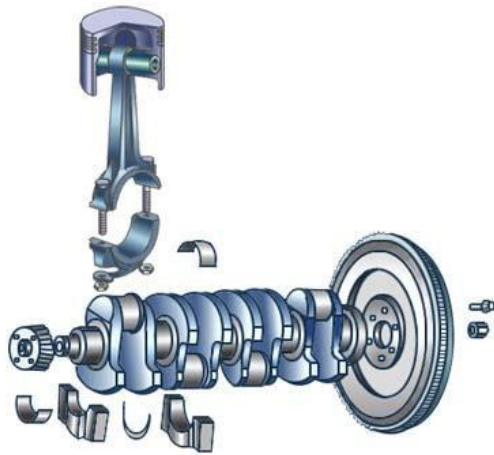
Two-stroke engines have four main disadvantage:

1. The lack of a dedicated lubrication system means that the parts of a two-stroke engine **wear** a lot faster.
2. Two-stroke **oil** is expensive, and you need about 4 ounces of it per gallon of gas to mix with fuel for lubricating parts. You would burn about a gallon of oil every 1,000 miles if you used a two-stroke engine in a car.
3. Two-stroke engines do not use **fuel** efficiently because some of mixture will directly leak out from exhaust post.
4. Two-stroke engines produce a lot of **pollution** because of burnt oil and leaked mixture.

第四节 发动机的总体构造

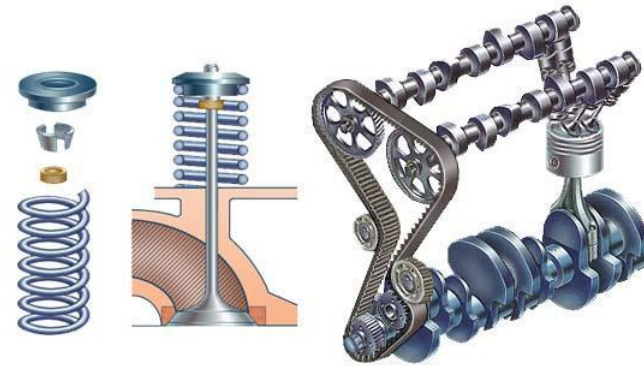
The Overall Structure of Engines

- 机体组 Engine Block
- 曲柄连杆机构 Crank and Connecting Rod
- 配气机构 Valve Train
- 供给系 Fuel and Air Supply System
- 点火系 Ignition System
- 冷却系 Cooling System
- 润滑系 Lubrication System
- 起动系 Starting System



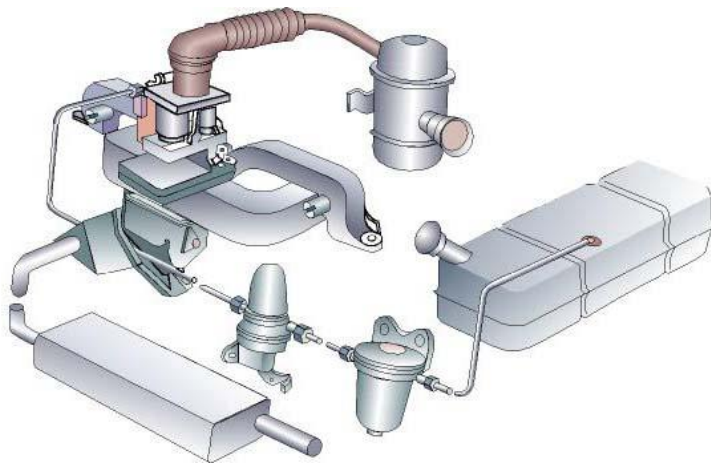
曲柄连杆机构

Crank and Connecting Rod



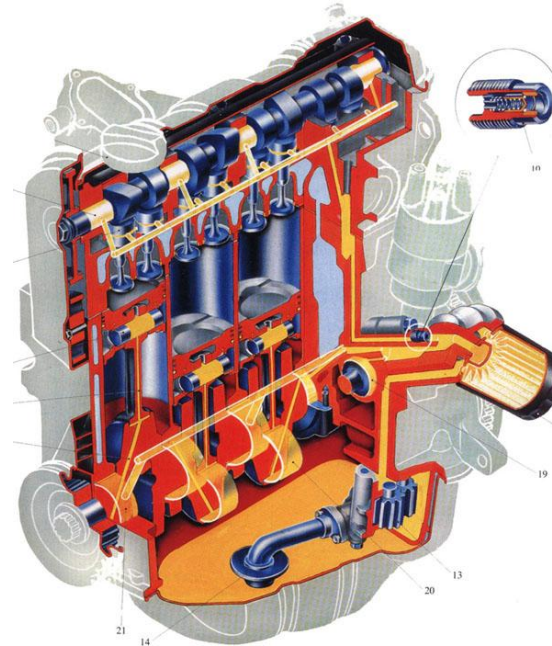
配气机构

Valve Train

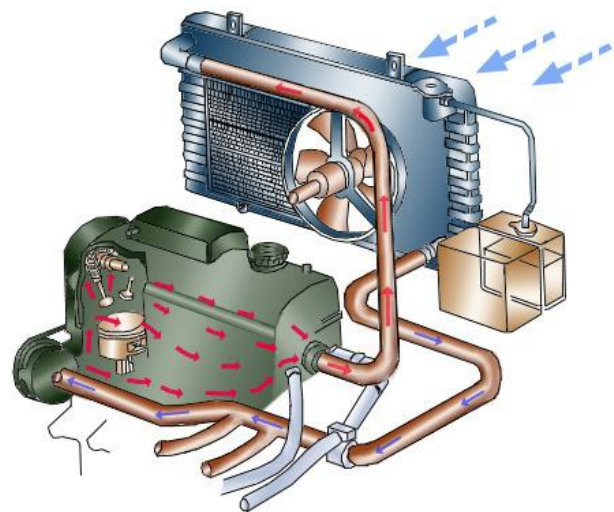


燃油供给系

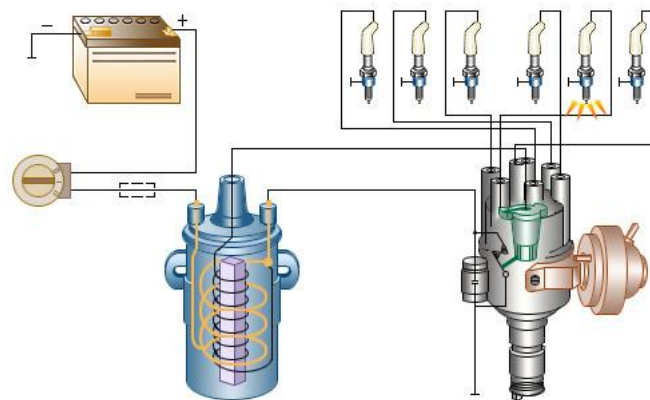
Fuel Supply System



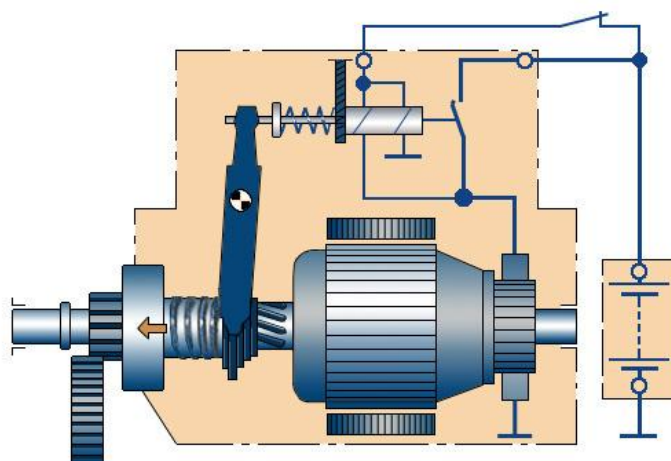
润滑系
Lubrication
System



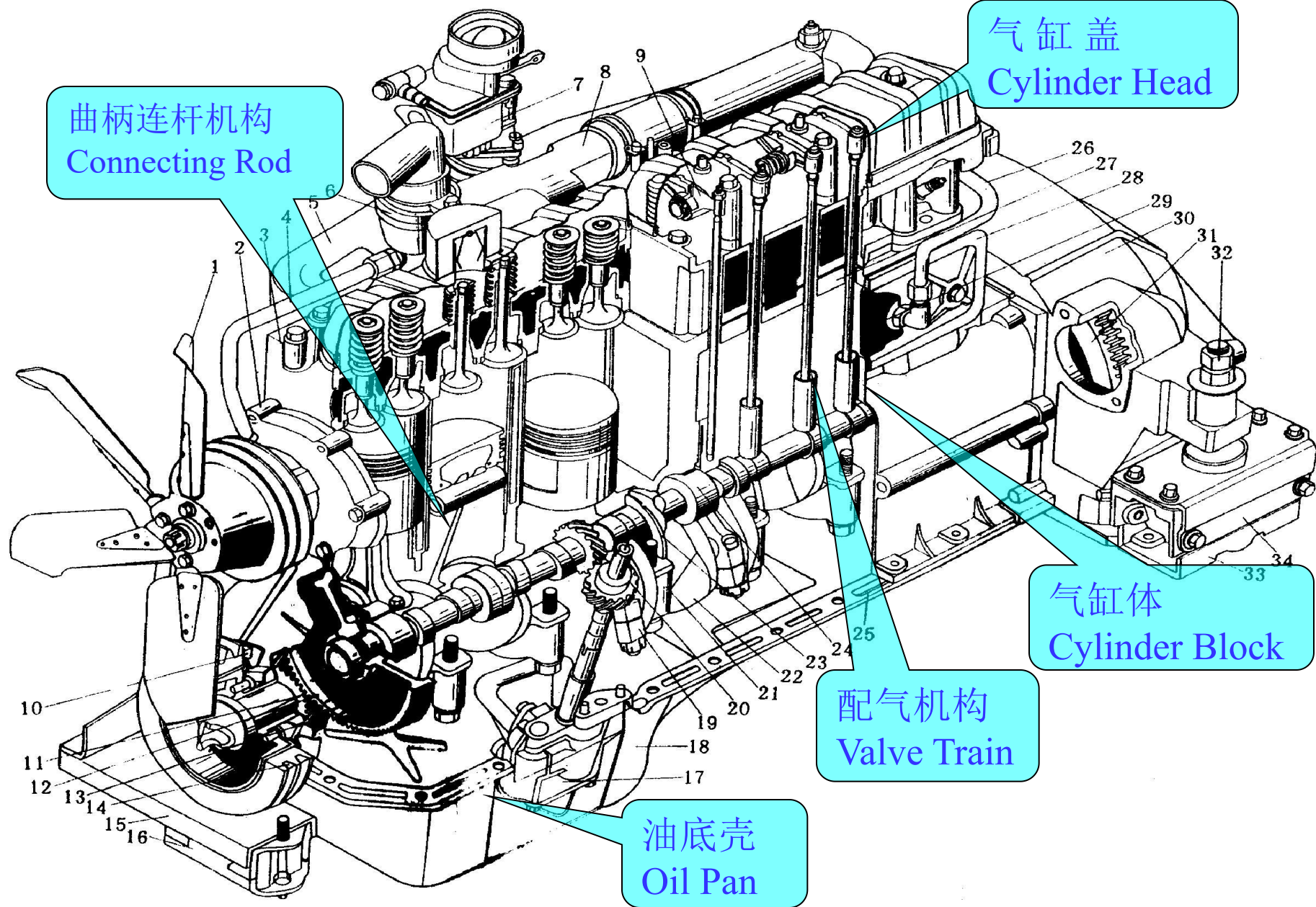
冷却系统
Cooling System

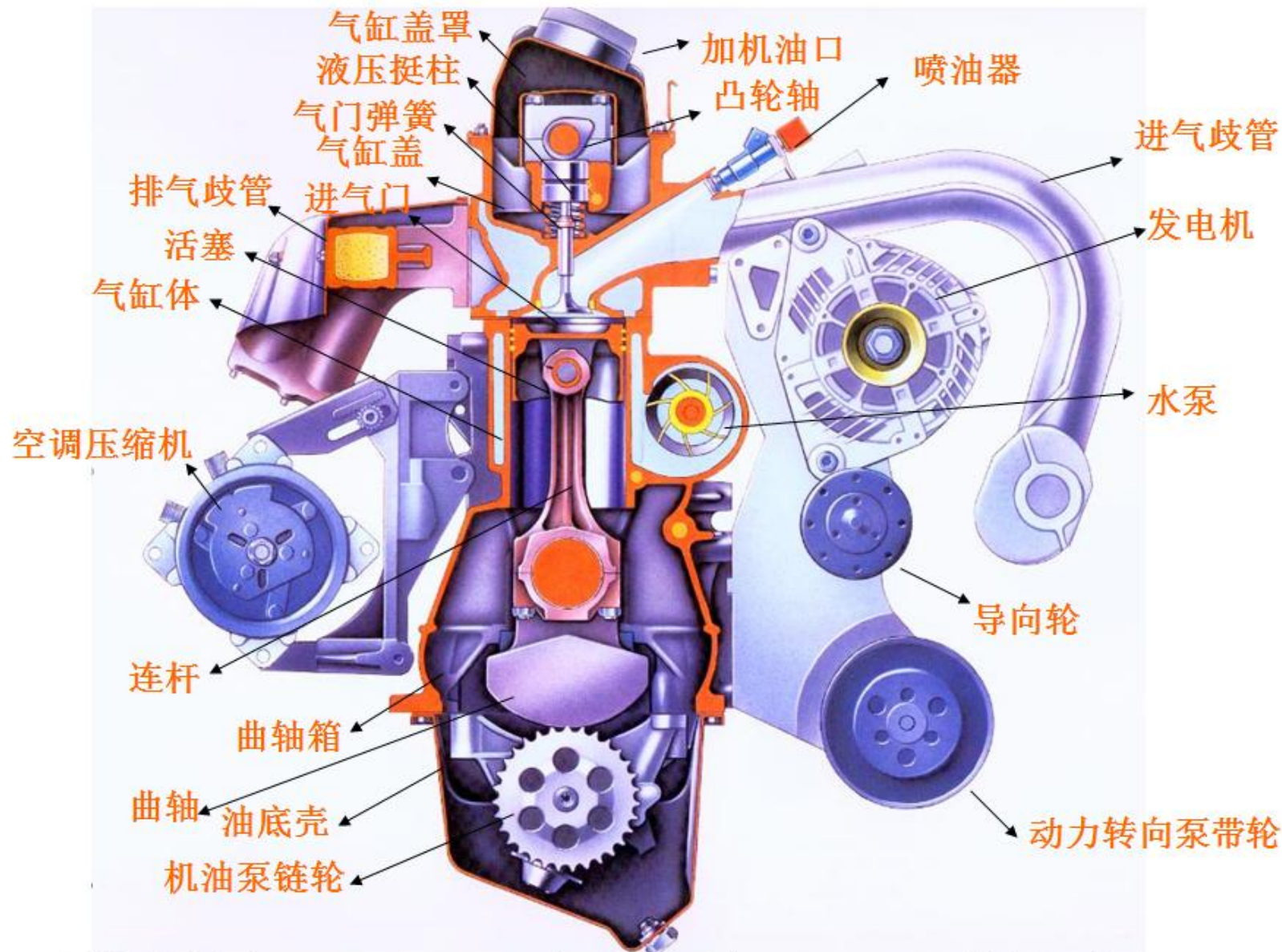


点火系统
Ignition System



起动系统
Starting System





桑塔纳轿车（2000GSI）电喷发动机（AJR）横剖面图



第五节 发动机的性能指标

Engine Operating Performance Specifications

发动机的性能指标是用来衡量发动机性能好坏的标准。发动机的主要性能指标有：动力性能指标，经济性能指标和排放性能指标。

The performance specifications of engine are generally used to present engine performances. Main engine performance specifications include : power output, fuel consumption and engine out emissions.

(1) 动力性能指标：动力性能指标指曲轴对外作功能力的指标，包括有效转矩 T_e 、有效功率 P_e 和曲轴转速 n 。

(1) Power output specification: Power output refers to the ability of crankshaft making useful work, including the effective torque T_e , the effective power P_e and crankshaft speed n .

- **有效转矩：** 发动机通过飞轮对外输出的转矩 T_e (N · m) 。有效转矩与外界施加于发动机曲轴上的阻力矩相平衡。
- **The effective torque :** the torque T_e (N · m) is that the engine outputs through the flywheel . Effective torque equals with the drag torque applied to the crankshaft from outside.
- **有效功率：** 发动机通过飞轮对外输出的功率 P_e (kW) 。等于有效转矩与曲轴角速度的乘积，可以用台架试验方法测定：
- **The effective power:** the power P_e (kW) that the engine outputs through the flywheel . It is equal to the product of the effective torque and crankshaft angular velocity . It can be determined by bench test.

$$P_e = T_e \cdot \frac{2 \pi \cdot n}{60} \times 10^{-3} = \frac{T_e \cdot n}{9550} \quad (\text{kW})$$

T_e —有效转矩，N · m

n —曲轴转速，r / min

- **额定功率/额定转速：** 发动机产品铭牌上标明的功率及相应转速称为额定功率和额定转速。
- **Brake power / brake speed :** power and corresponding speed marked on the engine nameplate are called the rated power and rated speed .

(2) 经济性能指标：通常用燃油消耗率来评价内燃机的经济性能。

(2) Fuel Economy: usually to evaluate the economic performance of the internal combustion engine with the fuel consumption rate .

- **燃油消耗率**：发动机每发出1kW有效功率，在1h内所消耗的燃油质量（以g为单位） g_e ，燃油消耗率越低，经济性越好。
- **Fuel consumption rate** refers to the fuel consumption of the unit effective work , that is, the engine outputting 1kW effective power consumes the fuel mass (in g) within 1 hour. Lower fuel consumption means the better economy.

$$g_e = \frac{G_f}{P_e} \bullet 10^3 \quad (g / kW \bullet h)$$

G_f —发动机每单位时间的耗油量(Fuel consumption in unit time), kg / h
 P_e —发动机的有效功率(The effective power), kW。

(3) 排放性能指标： 排放性能指标包括排放烟度、有害气体(CO, HC, NO_x)排放量、 噪声等。

(3) Emissions Levels: emissions performance configures include particles(PM), harmful gases (CO , HC, NO_x) emissions , noise and CO₂ .

发动机特性：发动机的性能随着许多因素而变化的规律。

Engine Characteristics : the law in which the engine performance changes with many factors.

- **发动机速度特性：**发动机的功率，转矩和燃油消耗率三者随曲轴转速变化的规律。
- **Engine speed characteristics :** the law in which engine power , torque and fuel consumption rate change with the crankshaft speed

有效功率测试 (The test of effective power)

发动机在试验台上(例如测功器试验台)进行试验而求得。

We can obtain the effective power by the test bench.

油耗测试 (Fuel consumption test)

可测出消耗一定量汽油所经历的时间，用以换算出发动机每小时油耗量Gf，计算出燃油消耗率ge。

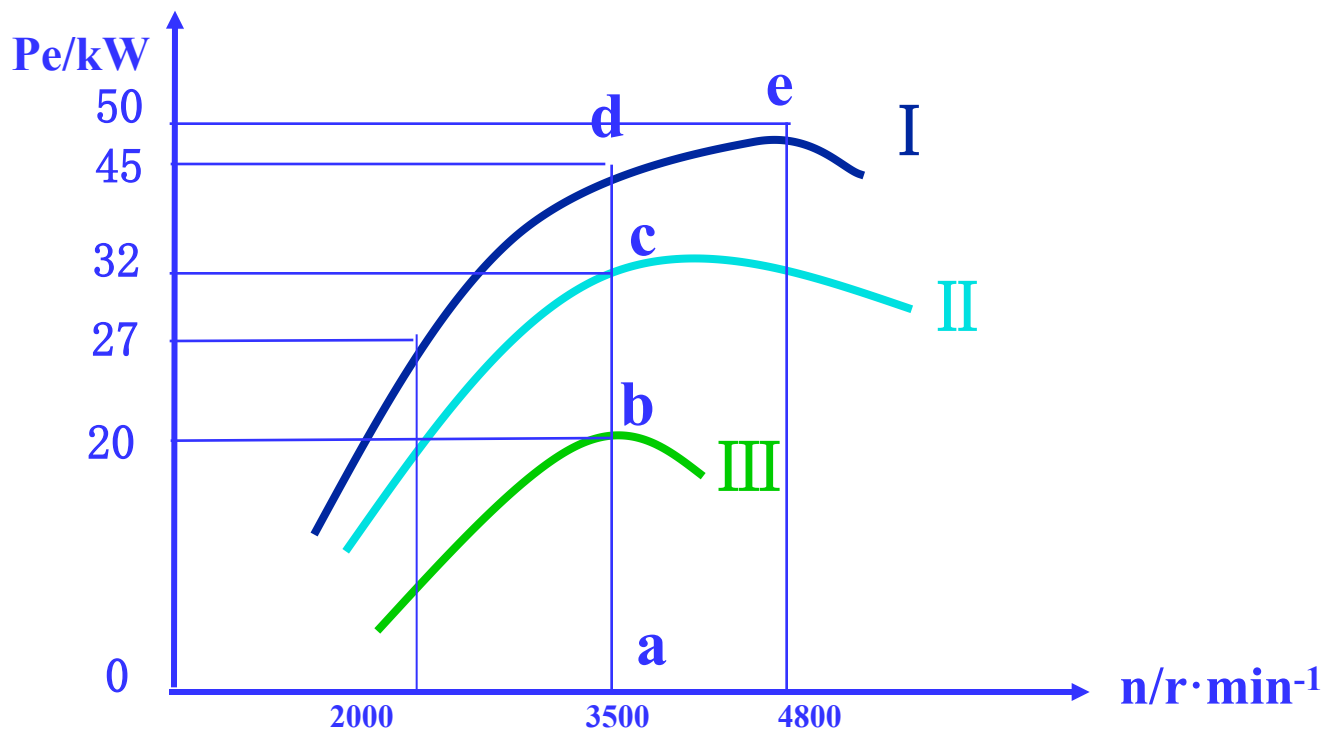
工况（指发动机的工作状况） **Working Condition**

一般用它的功率与曲轴转速来表征，有时也用负荷与曲轴转速来表明。

负荷 **Load**

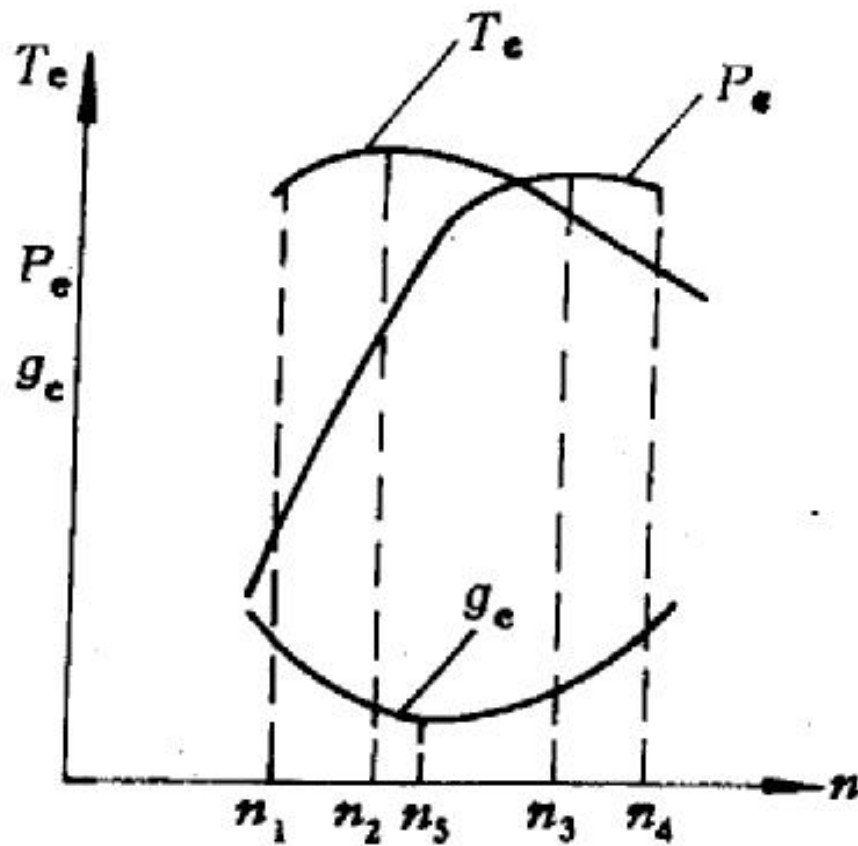
发动机在某一转速下的负荷，就是当时发动机发出的功率与同一转速下所可能发出的最大功率之比。

分析
右图
发动
机的
负荷



发动机全负荷速度特性（又称为发动机外特性）

Full Load Characteristic / External Characteristic



产品命名

